

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC Combined Geo-Scientist

Subject: Geophysics

Questions: 278

Source: Official source PDF extraction

AI-generated PYQs: 0

1. [UNKNOWN] paper-ii

1. Bode's law is associated with: (a) Distance between two planets (b) Distance between sun and planets (c) Distance between earth and moon (d) Distance between two stars

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

2. [UNKNOWN] paper-ii

1. Pulling force of a sinking lithospheric slab

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

3. [UNKNOWN] paper-ii

1. The time difference of S wave and P wave determines the epicentral angle to the source

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4. [UNKNOWN] paper-ii

1. It is not linked with the motion of charges

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5. [UNKNOWN] paper-ii

1. The tangential component of electric field is zero along the perfectly conducting surface

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6. [UNKNOWN] paper-ii

1. The field vectors and are not in phase; but lags behind

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8. [UNKNOWN] paper-ii

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13. [UNKNOWN] paper-ii

2. Which one of the following is NOT a branch of geophysics? (a) Plutology (b) Seismology (c) Hydrology (d) Topographic surveying and mapping

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14. [UNKNOWN] paper-ii

2. Changes in the convection of the core

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

15. [UNKNOWN] paper-ii

2. Pushing force of sliding off a mid oceanic ridge

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16. [UNKNOWN] paper-ii

2. The time difference of S wave and PP wave determines the depth of the hypo centre

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17. [UNKNOWN] paper-ii

2. Gauss law for free ions embedded in dielectric material follows which of the above equations? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

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18. [UNKNOWN] paper-ii

2. It has finite value even in a perfect vacuum Which of the statements given above is/are correct? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

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19. [UNKNOWN] paper-ii

2. The boundary condition for electric field vector is found from the fact that no work is done in taking a unit test charge around a closed path Which of the above statements is/are correct? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

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20. [UNKNOWN] paper-ii

2. The field vectors and are not in phase; but lags behind Which of the above statement is/are correct? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2 ===== PAGE 15 =====

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21. [UNKNOWN] paper-ii

2. Which one of the following is NOT a branch of geophysics? (a) Plutology (b) Seismology (c) Hydrology (d) Topographic surveying and mapping

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29. [UNKNOWN] paper-ii

3. Which branch of geophysics deals with the internal structure of earth? (a) Active reflection seismics (b) Active refraction seismics (c) Passive seismology (d) Magnetotellurics

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30. [UNKNOWN] paper-ii

3. Large earthquakes Select the correct answer using the code given below: (a) 1 and 3 only (b) 1 and 2 only (c) 2 only (d) 1, 2 and 3

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31. [UNKNOWN] paper-ii

3. Rising convection current at boundary regions Select the correct answer using the code given below: (a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3 ===== PAGE 5 =====

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32. [UNKNOWN] paper-ii

3. P-S arrival time difference from two stations determines the location of the earthquake Select the correct answer using the code given below: (a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

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33. [UNKNOWN] paper-ii

3. Gravitational force Select the correct answer using the code given below: (a) 1 only (b) 2 and 3 only (c) 1 and 2 only

(d) 1, 2 and 3 ===== PAGE 7 =====

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34. [UNKNOWN] paper-ii

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39. [UNKNOWN] paper-ii

4. Which one amongst the following phenomena provides evidence that 'the universe is expanding'? (a) Newton's law of gravitation (b) Coloumb's law (c) Green house effect (d) Doppler effect

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40. [UNKNOWN] paper-ii

4. Anti-ferromagnetic Code: A B C D (a) 1 3 4 2 (b) 1 4 3 2 (c) 2 3 4 1 (d) 2 4 3 1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

41. [UNKNOWN] paper-ii

4. Strangeness Select the correct answer using the code given below: (a) 1 and 2 only (b) 1, 2 and 3 only (c) 3 and 4 only (d) 1, 2, 3 and 4

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

42. [UNKNOWN] paper-ii

4. $N = M$ and $K = M$ Select the correct answer using the code given below: (a) 1 and 4 (b) 2 and 3 (c) 1 and 3 (d) 2 and 4

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43. [UNKNOWN] paper-ii

4. Porosity Select the correct answer using the code given below: (a) 1, 2, 3 and 4 (b) 1 and 4 only (c) 3 and 4 only (d) 1, 2 and 3 only

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44. [UNKNOWN] paper-ii

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45. [UNKNOWN] paper-ii

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46. [UNKNOWN] paper-ii

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47. [UNKNOWN] paper-ii

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48. [UNKNOWN] paper-ii

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49. [UNKNOWN] paper-ii

5. A three component seismometer was installed at a granite site. The thickness of rock layer is 15 km. An earthquake occurred at a depth of 10 km with an epicentral distance of 30 km. The observed seismic moment is 6×10^{27} . Which one of the following is the correct relationship among body-wave magnitude (m_b), surface-wave magnitude (M_s) and moment magnitude (M_w)? (a) $m_b < M_s = M_w$ (b) $m_b = M_s < M_w$ (c) $m_b < M_s < M_w$ (d) $m_b > M_s > M_w$ ===== PAGE 2 =====

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50. [UNKNOWN] paper-ii

5. A three component seismometer was installed at a granite site. The thickness of rock layer is 15 km. An earthquake occurred at a depth of 10 km with an epicentral distance of 30 km. The observed seismic moment is 6×10^{27} . Which one of the following is the correct relationship among body-wave magnitude (m_b), surface-wave magnitude (M_s) and moment magnitude (M_w)? (a) $m_b < M_s = M_w$ (b) $m_b = M_s < M_w$ (c) $m_b < M_s < M_w$ (d) $m_b > M_s > M_w$ ===== PAGE 2 =====

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51. [UNKNOWN] paper-ii

6. The core of Jupiter is surrounded by the layers of: (a) Liquid hydrogen and iron (b) Liquid-metallic, non-metallic and gaseous hydrogen (c) Gaseous hydrogen and gaseous helium only (d) Gaseous hydrogen, gaseous helium and liquid helium

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52. [UNKNOWN] paper-ii

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53. [UNKNOWN] paper-ii

7. The Earth's magnetic poles are shifting because of:

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54. [UNKNOWN] paper-ii

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55. [UNKNOWN] paper-ii

8. Acceleration due to gravity at a distance ($r > R_{\text{earth}}$) from the centre of earth is given by: (a) $+ = 0$ (b) $- = 0$ (c) $- = 0$ (d) $+ = 0$

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56. [UNKNOWN] paper-ii

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57. [UNKNOWN] paper-ii

9. Aurora Borealis occurs near the: (a) Magnetic north pole of earth (b) Equator of earth (c) Sea shore (d) Desert region

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58. [UNKNOWN] paper-ii

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59. [UNKNOWN] paper-ii

10. The distance between south geographic pole and south magnetic pole is approximately: (a) 4000 km (b) 1500 km (c) 1020 km (d) 2860 km ===== PAGE 3 =====

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60. [UNKNOWN] paper-ii

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61. [UNKNOWN] paper-ii

11. At neutral point of a bar magnet, a compass needle can stay in any position because: (a) Magnetic declination at that point is zero (b) Angle of dip at that point is zero (c) Earth's magnetic field is zero (d) Magnetic field of the bar magnet cancels the horizontal component of earth's magnetic field

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62. [UNKNOWN] paper-ii

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63. [UNKNOWN] paper-ii

12. Match List-I with List-II and select the correct answer using the code given below: List-I List-II (Mineral) (Magnetic property) A. Calcite

64. [UNKNOWN] paper-ii

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65. [UNKNOWN] paper-ii

13. Which of the following has maximum Koenigsberger ratio? (a) Igneous rock (b) Sedimentary rock (c) Metamorphic rock (d) Depends only on temperature of rock

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66. [UNKNOWN] paper-ii

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67. [UNKNOWN] paper-ii

14. Which one of the following is believed to be the cause of earth's magnetism? (a) The earth has a large bar magnet at its centre (b) The abundance of magnetic materials present in the crust of earth (c) The electric currents conducting in the liquid core (d) The magnetic fields produced by the magnetic materials present in the solid mantle ===== PAGE 4 =====

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68. [UNKNOWN] paper-ii

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69. [UNKNOWN] paper-ii

15. When an elastic wave propagates in a medium, its velocity depends on: (a) Density of the medium only (b) Elasticity of the medium only (c) Both density and elasticity of the medium (d) Neither density nor elasticity of the medium

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70. [UNKNOWN] paper-ii

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71. [UNKNOWN] paper-ii

16. In travel time versus distance plot in seismic refraction, the slope of the segments depends on the: (a) Impedance contrast between two layers (b) Seismic velocity and dip in each layer (c) Seismic velocity in each layer (d) Thickness of each layer

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72. [UNKNOWN] paper-ii

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73. [UNKNOWN] paper-ii

17. Which one of the following statements regarding planet earth is NOT correct? (a) Its shape is oblate spheroid (b) Its polar radius is shorter than its equatorial radius (c) Some of the oceanic islands are formed by the basalt volcanoes (d) The deepest layer of the earth is core which is composed of liquid only

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74. [UNKNOWN] paper-ii

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75. [UNKNOWN] paper-ii

18. The most homogeneous layer of the earth is: (a) Oceanic crust (b) Mantle (c) Outer core (d) Inner core

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76. [UNKNOWN] paper-ii

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77. [UNKNOWN] paper-ii

19. Plate recycling occurs below the surface of earth because of:

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78. [UNKNOWN] paper-ii

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79. [UNKNOWN] paper-ii

20. Two seismic recording stations receive data of S wave, P wave and PP wave consequently. On that basis, which of the following statements are correct?

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80. [UNKNOWN] paper-ii

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81. [UNKNOWN] paper-ii

21. The power of an earthquake of magnitude 7 is higher than that of 5 on Richter scale by: (a) 2 times (b) $7/5$ times (c) 31 times (d) 961 times

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82. [UNKNOWN] paper-ii

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83. [UNKNOWN] paper-ii

22. The directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at a point P: (2, 1, 3) in the direction of $\nabla f = -2$ is: (a) (b) (c) (d)

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84. [UNKNOWN] paper-ii

22. The directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at a point P: (2, 1, 3) in the direction of $\nabla f = -2$ is: (a) (b) (c) (d)

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85. [UNKNOWN] paper-ii

23. The divergence of the vector $\sin y - \cos y + xy$ is: (a) $\sin y$ (b) $2 \cos y$ (c) xy (d) 0 (zero)

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86. [UNKNOWN] paper-ii

23. The divergence of the vector $\sin y - \cos y + xy$ is: (a) $\sin y$ (b) $2 \cos y$ (c) xy (d) 0 (zero)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

87. [UNKNOWN] paper-ii

24. The Eigen values of the matrix are: (a) 1, -1 (b) 1, -4 (c) -1, 3 (d) 1, -3 ===== PAGE 6 =====

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88. [UNKNOWN] paper-ii

24. The Eigen values of the matrix are: (a) 1, -1 (b) 1, -4 (c) -1, 3 (d) 1, -3 ===== PAGE 6 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

89. [UNKNOWN] paper-ii

25. Which one of the following matrices is an orthogonal matrix? (a) (b) (c) (d)

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90. [UNKNOWN] paper-ii

25. Which one of the following matrices is an orthogonal matrix? (a) (b) (c) (d)

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91. [UNKNOWN] paper-ii

26. In gravity method, the vertical fault block can be modeled as: (a) Horizontal cylinder (b) Vertical cylinder (c) Infinite horizontal slab (d) Semi-infinite horizontal slab

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

92. [UNKNOWN] paper-ii

26. In gravity method, the vertical fault block can be modeled as: (a) Horizontal cylinder (b) Vertical cylinder (c) Infinite horizontal slab (d) Semi-infinite horizontal slab

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

93. [UNKNOWN] paper-ii

27. If the gravitational potential on the surface of the earth is 6×10^{10} ergs/gm, then the gravitational potential at the center of the earth can be: (a) 6×10^{10} ergs/gm (b) 9×10^{10} ergs/gm (c) 3×10^{10} ergs/gm (d) 7.5×10^{10} ergs/gm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

94. [UNKNOWN] paper-ii

27. If the gravitational potential on the surface of the earth is 6×10^{10} ergs/gm, then the gravitational potential at the center of the earth can be: (a) 6×10^{10} ergs/gm (b) 9×10^{10} ergs/gm (c) 3×10^{10} ergs/gm (d) 7.5×10^{10} ergs/gm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

95. [UNKNOWN] paper-ii

28. Two spheres of equal mass are separated by a distance of 'r' and are subjected to their mutual gravitational attraction. Which of the following quantities must have the same magnitude for both the spheres?

96. [UNKNOWN] paper-ii

28. Two spheres of equal mass are separated by a distance of 'r' and are subjected to their mutual gravitational attraction. Which of the following quantities must have the same magnitude for both the spheres?

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

97. [UNKNOWN] paper-ii

29. Which one of the following systems shows an example of the presence of at least three fundamental forces? (a) Pair of neutrons (n-n) (b) Pair of electrons (e-e) (c) Pair of protons (p-p) (d) Neutron and proton (n-p)

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98. [UNKNOWN] paper-ii

29. Which one of the following systems shows an example of the presence of at least three fundamental forces? (a) Pair of neutrons (n-n) (b) Pair of electrons (e-e) (c) Pair of protons (p-p) (d) Neutron and proton (n-p)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

99. [UNKNOWN] paper-ii

30. Which of the following quantities are conserved in a weak interaction?

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100. [UNKNOWN] paper-ii

30. Which of the following quantities are conserved in a weak interaction?

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101. [UNKNOWN] paper-ii

31. α -decay and β -decay are due to: (a) strong force and weak force respectively (b) electromagnetic force and weak force respectively (c) electrostatic force and weak force respectively (d) electromagnetic force and strong force respectively

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102. [UNKNOWN] paper-ii

31. α -decay and β -decay are due to: (a) strong force and weak force respectively (b) electromagnetic force and weak force respectively (c) electrostatic force and weak force respectively (d) electromagnetic force and strong force respectively

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

103. [UNKNOWN] paper-ii

32. A sphere, a cylinder and ring of same radius are allowed to roll down simultaneously on an inclined plane from the same height without slipping. Which of the following will reach the bottom first? (a) Sphere (b) Cylinder (c) Ring (d) All will reach simultaneously

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

104. [UNKNOWN] paper-ii

32. A sphere, a cylinder and ring of same radius are allowed to roll down simultaneously on an inclined plane from the same height without slipping. Which of the following will reach the bottom first? (a) Sphere (b) Cylinder (c) Ring (d) All will reach simultaneously

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

105. [UNKNOWN] paper-ii

33. Two particles of equal mass travelling with velocities 2m/s and 4m/s collide and stick together. What would be the velocity of the combined mass? (a) 2m/s (b) 4m/s (c) 3m/s (d) 6m/s ===== PAGE 8 =====

106. [UNKNOWN] paper-ii

33. Two particles of equal mass travelling with velocities 2m/s and 4m/s collide and stick together. What would be the velocity of the combined mass? (a) 2m/s (b) 4m/s (c) 3m/s (d) 6m/s ===== PAGE 8 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

107. [UNKNOWN] paper-ii

34. A slender uniform rod of total length 'L' and total mass 'M' has its rotational axis at the center of the length and perpendicular to length. Its moment of inertia can be given by: (a) ML^2 (b) $\frac{1}{12}ML^2$ (c) $\frac{1}{2}ML^2$ (d) $\frac{1}{3}ML^2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

108. [UNKNOWN] paper-ii

34. A slender uniform rod of total length 'L' and total mass 'M' has its rotational axis at the center of the length and perpendicular to length. Its moment of inertia can be given by: (a) ML^2 (b) $\frac{1}{12}ML^2$ (c) $\frac{1}{2}ML^2$ (d) $\frac{1}{3}ML^2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

109. [UNKNOWN] paper-ii

35. Total number of degrees of freedom for a rigid body around a fixed point is: (a) Zero (b) One (c) Two (d) Three

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

110. [UNKNOWN] paper-ii

35. Total number of degrees of freedom for a rigid body around a fixed point is: (a) Zero (b) One (c) Two (d) Three

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

111. [UNKNOWN] paper-ii

36. The speed of an electron in a uniform electric field changes from $0.95c$ to $0.98c$, where c is the velocity of light. Change in the mass of electron will be: (Rest mass of electron = 9.11×10^{-31} kg) (a) 1.8×10^{-31} kg (b) 16.4×10^{-27} kg (c) 1.8×10^{-27} kg (d) 16.4×10^{-31} kg

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

112. [UNKNOWN] paper-ii

36. The speed of an electron in a uniform electric field changes from $0.95c$ to $0.98c$, where c is the velocity of light. Change in the mass of electron will be: (Rest mass of electron = 9.11×10^{-31} kg) (a) 1.8×10^{-31} kg (b) 16.4×10^{-27} kg (c) 1.8×10^{-27} kg (d) 16.4×10^{-31} kg

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

113. [UNKNOWN] paper-ii

37. What will be the speed of a proton, whose kinetic energy is double to its rest mass energy? (a) c (b) $\frac{c}{2}$ (c) $\frac{c}{\sqrt{2}}$ (d) $\frac{c}{\sqrt{3}}$ ===== PAGE 9 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

114. [UNKNOWN] paper-ii

37. What will be the speed of a proton, whose kinetic energy is double to its rest mass energy? (a) c (b) $\frac{c}{2}$ (c) $\frac{c}{\sqrt{2}}$ (d) $\frac{c}{\sqrt{3}}$ ===== PAGE 9 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

115. [UNKNOWN] paper-ii

38. In geophysical inverse problem, if the rank of a matrix is K , N is the number of data points and M is the number of unknown parameters, then the condition for over determined problem is:

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116. [UNKNOWN] paper-ii

38. In geophysical inverse problem, if the rank of a matrix is K , N is the number of data points and M is the number of unknown parameters, then the condition for over determined problem is:

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

117. [UNKNOWN] paper-ii

39. Which one of the following distributions is symmetric about its mean and has zero skewness? (a) Laplacian distribution (b) Gaussian distribution (c) Poisson distribution (d) Binomial distribution

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118. [UNKNOWN] paper-ii

39. Which one of the following distributions is symmetric about its mean and has zero skewness? (a) Laplacian distribution (b) Gaussian distribution (c) Poisson distribution (d) Binomial distribution

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

119. [UNKNOWN] paper-ii

40. The probability of having at least two tails in 4 throws with a fair coin is: (a) (b) (c) (d)

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120. [UNKNOWN] paper-ii

40. The probability of having at least two tails in 4 throws with a fair coin is: (a) (b) (c) (d)

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121. [UNKNOWN] paper-ii

41. Characteristic polynomial of the matrix A is: (a) $t^2 - 5t + 1$ (b) $t^2 + 9$ (c) $t^2 - 3t + 7$ (d) $t^2 - 3t - 18$

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122. [UNKNOWN] paper-ii

41. Characteristic polynomial of the matrix A is: (a) $t^2 - 5t + 1$ (b) $t^2 + 9$ (c) $t^2 - 3t + 7$ (d) $t^2 - 3t - 18$

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

123. [UNKNOWN] paper-ii

42. The two vectors $U = (1, -3)$ and $V = (-2, 6)$ are: (a) Linearly dependent (b) Linearly independent (c) U is subset of V (d) V is subset of U ===== PAGE 10 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

124. [UNKNOWN] paper-ii

42. The two vectors $U = (1, -3)$ and $V = (-2, 6)$ are: (a) Linearly dependent (b) Linearly independent (c) U is subset of V (d) V is subset of U ===== PAGE 10 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

125. [UNKNOWN] paper-ii

43. The solution of the differential equation $y'' + 8y' + 15y = 0$ is: (a) $y = C_1 e^{3x} + C_2 e^{5x}$ (b) $y = C_1 e^{3x} + C_2 e^{3x}$ (c) $y = C_1 3x + C_2 5x$ (d) $y = C_1 e^{-3x} + C_2 e^{-5x}$

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

126. [UNKNOWN] paper-ii

43. The solution of the differential equation $y'' + 8y' + 15y = 0$ is: (a) $y = C_1 e^{3x} + C_2 e^{5x}$ (b) $y = C_1 e^{3x} + C_2 e^{3x}$ (c) $y = C_1 3x + C_2 5x$ (d) $y = C_1 e^{-3x} + C_2 e^{-5x}$

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

127. [UNKNOWN] paper-ii

44. In Laplace equation $\nabla^2 = 0$, all the stationary points of must belong to: (a) Maxima (b) Minima (c) Saddle points (d) Zero

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128. [UNKNOWN] paper-ii

44. In Laplace equation $\nabla^2 = 0$, all the stationary points of must belong to: (a) Maxima (b) Minima (c) Saddle points (d) Zero

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

129. [UNKNOWN] paper-ii

45. Interpolation is a method to: (a) Solve differential equations numerically (b) Find roots of a polynomial (c) Find the value beyond the points in a data set (d) Find the value of a point lying between known points

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130. [UNKNOWN] paper-ii

45. Interpolation is a method to: (a) Solve differential equations numerically (b) Find roots of a polynomial (c) Find the value beyond the points in a data set (d) Find the value of a point lying between known points

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

131. [UNKNOWN] paper-ii

46. We can find the value of the integral by using Simpson's 3/8 rule, if the interval number is: (a) 7 (b) 6 (c) 8 (d) any positive integer

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132. [UNKNOWN] paper-ii

46. We can find the value of the integral by using Simpson's 3/8 rule, if the interval number is: (a) 7 (b) 6 (c) 8 (d) any positive integer

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

133. [UNKNOWN] paper-ii

47. What is the magnetic force per unit length between two parallel wires separated by a distance 'd' each carrying a current 'I' in the same direction? (a) (b) (c) (d) ===== PAGE 11 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

134. [UNKNOWN] paper-ii

47. What is the magnetic force per unit length between two parallel wires separated by a distance 'd' each carrying a current 'I' in the same direction? (a) (b) (c) (d) ===== PAGE 11 =====

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135. [UNKNOWN] paper-ii

48. Two charges +2C and +3C are placed at the end points of a straight line AB. The point P at which the electric field is zero, when P lies in between A and B is: (Given AB = 4m) (a) at a distance 1.8 m from point B (b) at a distance 1.8 m from point A (c) at a distance 2.5 m from point B (d) at a distance 2.5 m from point A

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

136. [UNKNOWN] paper-ii

48. Two charges +2C and +3C are placed at the end points of a straight line AB. The point P at which the electric field is zero, when P lies in between A and B is: (Given AB = 4m) (a) at a distance 1.8 m from point B (b) at a distance 1.8 m from point A (c) at a distance 2.5 m from point B (d) at a distance 2.5 m from point A

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137. [UNKNOWN] paper-ii

49. A thin disc of radius 'R' carrying a uniform surface charge density ' σ ' rotates with a constant angular velocity ' ω ' about its axis (z-axis). The torque on the disc if placed in a uniform magnetic field acting along x-direction would be: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

138. [UNKNOWN] paper-ii

49. A thin disc of radius 'R' carrying a uniform surface charge density ' σ ' rotates with a constant angular velocity ' ω ' about its axis (z-axis). The torque on the disc if placed in a uniform magnetic field acting along x-direction would be: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

139. [UNKNOWN] paper-ii

50. The electrostatic force between two charged particles separated by a distance 'd' and placed in a medium having dielectric constant 'K' is 8 Newton. If the medium is replaced by another medium having dielectric constant '2K' and in the new medium, if the distance between two charged particles is increased to '2d', then the electrostatic force becomes: (a) 1 Newton (b) 2 Newton (c) 4 Newton (d) 8 Newton

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

140. [UNKNOWN] paper-ii

50. The electrostatic force between two charged particles separated by a distance 'd' and placed in a medium having dielectric constant 'K' is 8 Newton. If the medium is replaced by another medium having dielectric constant '2K' and in the new medium, if the distance between two charged particles is increased to '2d', then the electrostatic force becomes: (a) 1 Newton (b) 2 Newton (c) 4 Newton (d) 8 Newton

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141. [UNKNOWN] paper-ii

51. If in a cyclotron, a particle of charge 'q' and mass 'm' with a speed 'v' follows a circular path of radius 'R', because of a centripetal force 'F' exerted upon it by a magnetic field 'B', the radius of the path may be determined by using the equation: (a) (b) (c) (d) ===== PAGE 12 =====

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142. [UNKNOWN] paper-ii

51. If in a cyclotron, a particle of charge 'q' and mass 'm' with a speed 'v' follows a circular path of radius 'R', because of a centripetal force 'F' exerted upon it by a magnetic field 'B', the radius of the path may be determined by using the equation: (a) (b) (c) (d) ===== PAGE 12 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

143. [UNKNOWN] paper-ii

52. If the potential in a certain medium is given by where ' k ' is a constant, the total charge associated with the medium is: (a) q (b) 2q (c) 0 (zero) (d) q/ϵ_0

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

144. [UNKNOWN] paper-ii

52. If the potential in a certain medium is given by where ' k ' is a constant, the total charge associated with the medium is: (a) q (b) 2q (c) 0 (zero) (d) q/ϵ_0

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

145. [UNKNOWN] paper-ii

54. The solution of the differential equation with the boundary conditions $V(2) = 3$ and $V(5) = 0$ is: (a) $-x+5$ (b) $-x-5$ (c) $x+5$ (d) $x-5$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

146. [UNKNOWN] paper-ii

54. The solution of the differential equation with the boundary conditions $V(2) = 3$ and $V(5) = 0$ is: (a) $-x+5$ (b) $-x-5$ (c) $x+5$ (d) $x-5$

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

147. [UNKNOWN] paper-ii

55. A circular loop of wire of radius 'a' and resistance 'R' is oriented in the xy-plane. A uniform magnetic field of magnitude 'B' points in the + z-direction. If the loop of wire is rotated about the x-axis with an angular frequency ' ω ', what is the average power dissipated by the joule heating in the loop? (a) (b) (c) (d) 0 (zero) ===== PAGE 13 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

148. [UNKNOWN] paper-ii

55. A circular loop of wire of radius 'a' and resistance 'R' is oriented in the xy-plane. A uniform magnetic field of magnitude 'B' points in the + z-direction. If the loop of wire is rotated about the x-axis with an angular frequency ' ω ', what is the average power dissipated by the joule heating in the loop? (a) (b) (c) (d) 0 (zero) ===== PAGE 13 =====

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149. [UNKNOWN] paper-ii

56. Consider the following statements regarding displacement current:

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150. [UNKNOWN] paper-ii

56. Consider the following statements regarding displacement current:

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151. [UNKNOWN] paper-ii

57. Which one of the following statements is correct with reference to Faraday's law of electromagnetic induction? (a) Magnetic field produces electric field (b) Time varying electric field produces magnetic field (c) Time varying magnetic field produces conservative electric field (d) Time varying magnetic field produces non-conservative time varying electric field

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152. [UNKNOWN] paper-ii

57. Which one of the following statements is correct with reference to Faraday's law of electromagnetic induction? (a) Magnetic field produces electric field (b) Time varying electric field produces magnetic field (c) Time varying magnetic field produces conservative electric field (d) Time varying magnetic field produces non-conservative time varying electric field

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

153. [UNKNOWN] paper-ii

58. If a magnetic monopole existed, which one of the following expressions would be proportional to the "magnetic charge" of the monopole? (Assuming that there are no other sources of electric or magnetic fields present) (a) (b) (c) (d)

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

154. [UNKNOWN] paper-ii

58. If a magnetic monopole existed, which one of the following expressions would be proportional to the "magnetic charge" of the monopole? (Assuming that there are no other sources of electric or magnetic fields present) (a) (b) (c) (d)

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

155. [UNKNOWN] paper-ii

59. The magnetic field corresponding to the vector potential (where $\vec{a}$ is a constant vector) is: (a) (b) (c) (d)

156. [UNKNOWN] paper-ii

59. The magnetic field corresponding to the vector potential (where $\vec{A}$ is a constant vector) is: (a) $\vec{A}$ (b) $\vec{A} \times \vec{r}$ (c) $\vec{r} \times \vec{A}$ (d) $\vec{r} \cdot \vec{A}$

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157. [UNKNOWN] paper-ii

60. Which one of the following equations will remain the same while changing the medium? (a) $\vec{E} = -\nabla \phi$ (b) $\vec{B} = \nabla \times \vec{A}$ (c) $\vec{D} = \epsilon \vec{E}$ (d) $\vec{H} = \frac{1}{\mu} \nabla \times \vec{A}$ ===== PAGE 14 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

158. [UNKNOWN] paper-ii

60. Which one of the following equations will remain the same while changing the medium? (a) $\vec{E} = -\nabla \phi$ (b) $\vec{B} = \nabla \times \vec{A}$ (c) $\vec{D} = \epsilon \vec{E}$ (d) $\vec{H} = \frac{1}{\mu} \nabla \times \vec{A}$ ===== PAGE 14 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

159. [UNKNOWN] paper-ii

61. An electromagnetic wave crossing from one medium to other (if there is no free charge or free current at interface), then: (a) Parallel component of $\vec{E}$ and $\vec{B}$ are continuous (b) Perpendicular component of $\vec{E}$ and $\vec{B}$ are continuous (c) Perpendicular component of $\vec{E}$ and parallel component of $\vec{B}$ are continuous (d) Parallel component of $\vec{E}$ and perpendicular component of $\vec{B}$ are continuous

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

160. [UNKNOWN] paper-ii

61. An electromagnetic wave crossing from one medium to other (if there is no free charge or free current at interface), then: (a) Parallel component of $\vec{E}$ and $\vec{B}$ are continuous (b) Perpendicular component of $\vec{E}$ and $\vec{B}$ are continuous (c) Perpendicular component of $\vec{E}$ and parallel component of $\vec{B}$ are continuous (d) Parallel component of $\vec{E}$ and perpendicular component of $\vec{B}$ are continuous

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161. [UNKNOWN] paper-ii

62. Consider the following statements with regard to boundary condition:

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162. [UNKNOWN] paper-ii

62. Consider the following statements with regard to boundary condition:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

163. [UNKNOWN] paper-ii

63. Electromagnetic induction method of geophysics obeys: (a) Wave equation (b) Diffusion equation (c) Laplace equation (d) Poisson equation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

164. [UNKNOWN] paper-ii

63. Electromagnetic induction method of geophysics obeys: (a) Wave equation (b) Diffusion equation (c) Laplace equation (d) Poisson equation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

165. [UNKNOWN] paper-ii

64. Consider the following statements with regard to propagation of electromagnetic wave in a conducting medium:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

166. [UNKNOWN] paper-ii

64. Consider the following statements with regard to propagation of electromagnetic wave in a conducting medium:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

167. [UNKNOWN] paper-ii

65. If the electrical permittivity of certain non-magnetic medium is 140 times higher than the electrical permittivity of space, the speed of electromagnetic wave in this medium would be approximately: (a) 2142.9 km/s (b) 25354.6 km/s (c) 3×10^5 km/s (d) 3.6×10^5 km/s

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

168. [UNKNOWN] paper-ii

65. If the electrical permittivity of certain non-magnetic medium is 140 times higher than the electrical permittivity of space, the speed of electromagnetic wave in this medium would be approximately: (a) 2142.9 km/s (b) 25354.6 km/s (c) 3×10^5 km/s (d) 3.6×10^5 km/s

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

169. [UNKNOWN] paper-ii

66. Electromagnetic survey gives better results if the top surface layer is: (a) More conductive (b) Less conductive (c) Highly magnetic (d) Having same conductivity as sub-surface layers

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

170. [UNKNOWN] paper-ii

66. Electromagnetic survey gives better results if the top surface layer is: (a) More conductive (b) Less conductive (c) Highly magnetic (d) Having same conductivity as sub-surface layers

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

171. [UNKNOWN] paper-ii

67. In TURAM method an anomaly is indicated by: (a) Amplitude ratio of less than 1 and negative phase anomaly (b) Amplitude ratio of more than 1 and negative phase anomaly (c) Amplitude ratio of less than 1 and positive phase anomaly (d) Amplitude ratio of more than 1 and positive phase anomaly

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

172. [UNKNOWN] paper-ii

67. In TURAM method an anomaly is indicated by: (a) Amplitude ratio of less than 1 and negative phase anomaly (b) Amplitude ratio of more than 1 and negative phase anomaly (c) Amplitude ratio of less than 1 and positive phase anomaly (d) Amplitude ratio of more than 1 and positive phase anomaly

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

173. [UNKNOWN] paper-ii

68. In electrical resistivity methods, the coefficient of anisotropy () is generally: (a) Between 0 to 0.5 (b) Between 1 to 2 (c) More than 2 (d) Between 0.5 and 1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

174. [UNKNOWN] paper-ii

68. In electrical resistivity methods, the coefficient of anisotropy () is generally: (a) Between 0 to 0.5 (b) Between 1 to 2 (c) More than 2 (d) Between 0.5 and 1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

175. [UNKNOWN] paper-ii

69. The law of refraction for plane seismic waves can be written as: (a) = (b) = (c) = (d) = ===== PAGE 16 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

176. [UNKNOWN] paper-ii

69. The law of refraction for plane seismic waves can be written as: (a) $v_1 \sin i = v_2 \sin r$ (b) $v_1 \sin r = v_2 \sin i$ (c) $v_1 \sin i = v_2 \sin r$ (d) $v_1 \sin r = v_2 \sin i$ ===== PAGE 16 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

177. [UNKNOWN] paper-ii

70. The presence of gas in sedimentary rocks reduces:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

178. [UNKNOWN] paper-ii

70. The presence of gas in sedimentary rocks reduces:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

179. [UNKNOWN] paper-ii

71. If the acoustic impedances offered by two successive layers are same, then all the incident energy will be: (a) Reflected (b) Transmitted (c) Critically refracted (d) Partly reflected and partly transmitted

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

180. [UNKNOWN] paper-ii

71. If the acoustic impedances offered by two successive layers are same, then all the incident energy will be: (a) Reflected (b) Transmitted (c) Critically refracted (d) Partly reflected and partly transmitted

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

181. [UNKNOWN] paper-ii

72. A signal has a frequency spectrum of 0–5 kHz. If f_s is the sampling frequency of a data processing system then aliasing will arise when: (a) $f_s > 10$ kHz (b) $f_s = 10$ kHz (c) $f_s < 10$ kHz (d) $f_s = 20$ kHz

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

182. [UNKNOWN] paper-ii

72. A signal has a frequency spectrum of 0–5 kHz. If f_s is the sampling frequency of a data processing system then aliasing will arise when: (a) $f_s > 10$ kHz (b) $f_s = 10$ kHz (c) $f_s < 10$ kHz (d) $f_s = 20$ kHz

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

183. [UNKNOWN] paper-ii

73. The spectrum of a signal contains minimum and maximum frequency of 1kHz and 100 kHz respectively. To avoid aliasing effect, the signal must be sampled at: (a) more than 2 kHz (b) more than 200 kHz (c) more than 100 kHz (d) more than 1000 kHz

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

184. [UNKNOWN] paper-ii

73. The spectrum of a signal contains minimum and maximum frequency of 1kHz and 100 kHz respectively. To avoid aliasing effect, the signal must be sampled at: (a) more than 2 kHz (b) more than 200 kHz (c) more than 100 kHz (d) more than 1000 kHz

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

185. [UNKNOWN] paper-ii

74. In a seismic refraction survey, if V_u and V_d are updip and downdip velocity, T_u and T_d are the intercept times, then which one of the following relationships between velocities and intercept times is true? (a) $V_u > V_d$, $T_u < T_d$ (b) $V_u > V_d$, $T_u > T_d$ (c) $V_u < V_d$, $T_u < T_d$ (d) $V_u < V_d$, $T_u > T_d$ ===== PAGE 17 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

186. [UNKNOWN] paper-ii

74. In a seismic refraction survey, if V_u and V_d are updip and downdip velocity, T_u and T_d are the intercept times, then

which one of the following relationships between velocities and intercept times is true? (a) $V_u > V_d$, $T_u < T_d$ (b) $V_u > V_d$, $T_u > T_d$ (c) $V_u < V_d$, $T_u < T_d$ (d) $V_u < V_d$, $T_u > T_d$ ===== PAGE 17 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

187. [UNKNOWN] paper-ii

75. The input signal $I(t)$ and Output signal $g(t)$ are said to be linear, only if: (a) $g(t) I(t)$ (b) $g(t)$ (c) $g(t) = I(t) \cdot g(t)$ (d) $I(t) = g(t) \cdot I(t)$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

188. [UNKNOWN] paper-ii

75. The input signal $I(t)$ and Output signal $g(t)$ are said to be linear, only if: (a) $g(t) I(t)$ (b) $g(t)$ (c) $g(t) = I(t) \cdot g(t)$ (d) $I(t) = g(t) \cdot I(t)$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

189. [UNKNOWN] paper-ii

76. The Fourier transform of the function $f(t) = -$ is: (a) $\sin^3$ (b) $\cos^3$ (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

190. [UNKNOWN] paper-ii

76. The Fourier transform of the function $f(t) = -$ is: (a) $\sin^3$ (b) $\cos^3$ (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

191. [UNKNOWN] paper-ii

77. Transformation of a time domain signal to frequency domain is accomplished by: (a) Fourier transform (b) Laplace transform (c) Fourier series (d) Hilbert transform

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

192. [UNKNOWN] paper-ii

77. Transformation of a time domain signal to frequency domain is accomplished by: (a) Fourier transform (b) Laplace transform (c) Fourier series (d) Hilbert transform

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

193. [UNKNOWN] paper-ii

78. The two dimensional Fourier transform of a gravity data makes it possible to digitally filter the gravity anomalies. When the function representing gravity data is multiplied by the Fourier filter function, a new function is produced. This process is called: (a) De-convolution (b) Convolution (c) Digitization (d) Transformation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

194. [UNKNOWN] paper-ii

78. The two dimensional Fourier transform of a gravity data makes it possible to digitally filter the gravity anomalies. When the function representing gravity data is multiplied by the Fourier filter function, a new function is produced. This process is called: (a) De-convolution (b) Convolution (c) Digitization (d) Transformation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

195. [UNKNOWN] paper-ii

79. If the Laplace transform of is , then the Laplace transform of would be: (a) (b) (c) (d) ===== PAGE 18 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

196. [UNKNOWN] paper-ii

79. If the Laplace transform of is , then the Laplace transform of would be: (a) (b) (c) (d) ===== PAGE 18 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

197. [UNKNOWN] paper-ii

80. Relationship between Power Spectral Density (PSD) and autocorrelation of a real, stationary signal is: (a) PSD is the Fourier transform of autocorrelation (b) PSD is the Z transform of autocorrelation (c) PSD is the Laplace transform of autocorrelation (d) PSD is the Hilbert transform of autocorrelation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

198. [UNKNOWN] paper-ii

80. Relationship between Power Spectral Density (PSD) and autocorrelation of a real, stationary signal is: (a) PSD is the Fourier transform of autocorrelation (b) PSD is the Z transform of autocorrelation (c) PSD is the Laplace transform of autocorrelation (d) PSD is the Hilbert transform of autocorrelation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

199. [UNKNOWN] paper-ii

81. The Laplace transform of a function $f(t)$ is $L\{f(t)\} = F(s)$. Let $f(t) = e^{at}$, when t and ' a ' is a constant and ' a ' < s then the value of $L\{f(t)\}$ is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

200. [UNKNOWN] paper-ii

81. The Laplace transform of a function $f(t)$ is $L\{f(t)\} = F(s)$. Let $f(t) = e^{at}$, when t and ' a ' is a constant and ' a ' < s then the value of $L\{f(t)\}$ is: (a) (b) (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

201. [UNKNOWN] paper-ii

82. If two functions have lengths of 4 and 3 respectively then their convolved output has a length of: (a) 7 (b) 6 (c) 1 (d) -1

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202. [UNKNOWN] paper-ii

82. If two functions have lengths of 4 and 3 respectively then their convolved output has a length of: (a) 7 (b) 6 (c) 1 (d) -1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

203. [UNKNOWN] paper-ii

83. Autocorrelation function of a wave form contains information on: (a) Phase only (b) Phase and amplitude (c) Amplitude only (d) Noise

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

204. [UNKNOWN] paper-ii

83. Autocorrelation function of a wave form contains information on: (a) Phase only (b) Phase and amplitude (c) Amplitude only (d) Noise

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

205. [UNKNOWN] paper-ii

84. Cross correlation function measures the: (a) Degree of dissimilarity in wave forms (b) Degree of similarity in wave forms (c) Impulse response (d) De-convolution response

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

206. [UNKNOWN] paper-ii

84. Cross correlation function measures the: (a) Degree of dissimilarity in wave forms (b) Degree of similarity in wave forms (c) Impulse response (d) De-convolution response

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

207. [UNKNOWN] paper-ii

85. The value of integral $(2-4\cos^2)\cos$ is: (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{1}{4}$ (d) (zero) ===== PAGE 19 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

208. [UNKNOWN] paper-ii

85. The value of integral $(2-4\cos^2)\cos$ is: (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{1}{4}$ (d) (zero) ===== PAGE 19 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

209. [UNKNOWN] paper-ii

86. Consider the unit-step function $u(t-a)$. If $f(t)$ be a function then the product function $f(t)u(t-a)$ will produce: (a) Rotation effect (b) Damping effect (c) No change (d) Switching off and on effect

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210. [UNKNOWN] paper-ii

86. Consider the unit-step function $u(t-a)$. If $f(t)$ be a function then the product function $f(t)u(t-a)$ will produce: (a) Rotation effect (b) Damping effect (c) No change (d) Switching off and on effect

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

211. [UNKNOWN] paper-ii

87. The Laplace transform of a delta function at time $t = 1$ is: (a) 1 (b) e^{-s} (c) e^{-t} (d) $1/t$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

212. [UNKNOWN] paper-ii

87. The Laplace transform of a delta function at time $t = 1$ is: (a) 1 (b) e^{-s} (c) e^{-t} (d) $1/t$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

213. [UNKNOWN] paper-ii

88. The value of $\int_0^{\infty} e^{-st} \delta(t-1) dt$ will be: (a) 1 (b) 8 (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

214. [UNKNOWN] paper-ii

88. The value of $\int_0^{\infty} e^{-st} \delta(t-1) dt$ will be: (a) 1 (b) 8 (c) (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

215. [UNKNOWN] paper-ii

89. The ratio of total solar radiant energy returned by a body to the total radiant energy incident on the body is called: (a) Reflectance (b) Transmittance (c) Albedo (d) Reflection Coefficient

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

216. [UNKNOWN] paper-ii

89. The ratio of total solar radiant energy returned by a body to the total radiant energy incident on the body is called: (a) Reflectance (b) Transmittance (c) Albedo (d) Reflection Coefficient

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

217. [UNKNOWN] paper-ii

90. Which of the following is NOT used as principle of remote sensing? (a) Interaction of electromagnetic radiation with atmosphere (b) Interaction of electromagnetic radiation with earth surface (c) Interaction of electromagnetic radiation with satellites (d) Interaction of electromagnetic radiation with clouds and aerosols ===== PAGE 20 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

218. [UNKNOWN] paper-ii

90. Which of the following is NOT used as principle of remote sensing? (a) Interaction of electromagnetic radiation with atmosphere (b) Interaction of electromagnetic radiation with earth surface (c) Interaction of electromagnetic radiation with satellites (d) Interaction of electromagnetic radiation with clouds and aerosols ===== PAGE 20 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

219. [UNKNOWN] paper-ii

91. Which one of the following statement is correct for remote sensing techniques? (a) They use both sonic and electromagnetic waves for investigation (b) They use entire range (radio waves to gamma-rays) of the electromagnetic spectrum for investigation (c) They use microwave, infrared and visual portions of the electromagnetic spectrum for investigation (d) They gather the information with or without actual contact with the object or area under investigation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

220. [UNKNOWN] paper-ii

91. Which one of the following statement is correct for remote sensing techniques? (a) They use both sonic and electromagnetic waves for investigation (b) They use entire range (radio waves to gamma-rays) of the electromagnetic spectrum for investigation (c) They use microwave, infrared and visual portions of the electromagnetic spectrum for investigation (d) They gather the information with or without actual contact with the object or area under investigation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

221. [UNKNOWN] paper-ii

92. What is the refractive index of air at Standard Temperature Pressure (STP) for sodium yellow light? (a) Exactly 1.0 (b) About 0.03% higher than that of vacuum (c) About 0.03% lower than that of vacuum (d) It depends on the location

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

222. [UNKNOWN] paper-ii

92. What is the refractive index of air at Standard Temperature Pressure (STP) for sodium yellow light? (a) Exactly 1.0 (b) About 0.03% higher than that of vacuum (c) About 0.03% lower than that of vacuum (d) It depends on the location

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

223. [UNKNOWN] paper-ii

93. Which one of the following colours of light travel fastest through a transparent glass sheet? (a) Red (b) Yellow (c) Green (d) Violet

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

224. [UNKNOWN] paper-ii

93. Which one of the following colours of light travel fastest through a transparent glass sheet? (a) Red (b) Yellow (c) Green (d) Violet

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

225. [UNKNOWN] paper-ii

94. A photon and an electron have same De Broglie wavelength. Which one of the following is correct? (a) Electron moves faster than photon (b) Photon moves faster than electron (c) Both move with same speed (d) Cannot be predicted

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

226. [UNKNOWN] paper-ii

94. A photon and an electron have same De Broglie wavelength. Which one of the following is correct? (a) Electron moves faster than photon (b) Photon moves faster than electron (c) Both move with same speed (d) Cannot be predicted

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

227. [UNKNOWN] paper-ii

95. The planet Saturn has a surface temperature 80K. The wavelength corresponding to maximum radiation that can be detected from it is: (a) 330 nm (b) 330 Å (c) 33μm (d) 33mm ===== PAGE 21 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

228. [UNKNOWN] paper-ii

95. The planet Saturn has a surface temperature 80K. The wavelength corresponding to maximum radiation that can be detected from it is: (a) 330 nm (b) 330 Å (c) 33μm (d) 33mm ===== PAGE 21 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

229. [UNKNOWN] paper-ii

96. The total radiant emittance is given by $u = \sigma AT^4$. This equation represents: (a) Planck's radiation law (b) Wein's Law (c) Stefan-Boltzmann law (d) Kirchhoff's law

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

230. [UNKNOWN] paper-ii

96. The total radiant emittance is given by $u = \sigma AT^4$. This equation represents: (a) Planck's radiation law (b) Wein's Law (c) Stefan-Boltzmann law (d) Kirchhoff's law

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

231. [UNKNOWN] paper-ii

97. The surface temperature of the sun is 6000 K and it emits maximum energy at a wavelength of 4753 Å. If the moon emits maximum energy at a wavelength 15×10^{-6} m, then what would be the surface temperature of the moon? (a) 180 K (b) 190 K (c) 200 K (d) 210 K

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

232. [UNKNOWN] paper-ii

97. The surface temperature of the sun is 6000 K and it emits maximum energy at a wavelength of 4753 Å. If the moon emits maximum energy at a wavelength 15×10^{-6} m, then what would be the surface temperature of the moon? (a) 180 K (b) 190 K (c) 200 K (d) 210 K

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

233. [UNKNOWN] paper-ii

98. What is the most probable energy 'E' for photons emitted by a black body of temperature T, assuming that the intensity I (E) $\sim E^5 e^{-E/kT}$? (where k is Boltzmann constant) (a) kT (b) 2 kT (c) $\frac{1}{2}$ kT (d) 4 kT

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

234. [UNKNOWN] paper-ii

98. What is the most probable energy 'E' for photons emitted by a black body of temperature T, assuming that the intensity I (E) $\sim E^5 e^{-E/kT}$? (where k is Boltzmann constant) (a) kT (b) 2 kT (c) $\frac{1}{2}$ kT (d) 4 kT

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235. [UNKNOWN] paper-ii

99. The ratio of photons presents in 1cm³ of radiation in thermal equilibrium at 1000 K to the photons present in the same volume of radiation in thermal equilibrium at 8000 K is: (a) 1:2 (b) 2:1 (c) 1:64 (d) 1:512

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236. [UNKNOWN] paper-ii

99. The ratio of photons presents in 1cm³ of radiation in thermal equilibrium at 1000 K to the photons present in the same volume of radiation in thermal equilibrium at 8000 K is: (a) 1:2 (b) 2:1 (c) 1:64 (d) 1:512

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf

237. [UNKNOWN] paper-ii

100. Planck's formula for black body radiation reduces to Rayleigh and Wien's formulae respectively for: (a) Small and large wavelength (b) Large and small wavelength (c) Large and complex wavelength (d) Complex and large wavelength
===== PAGE 22 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf

238. [UNKNOWN] paper-ii

100. Planck's formula for black body radiation reduces to Rayleigh and Wien's formulae respectively for: (a) Small and large wavelength (b) Large and small wavelength (c) Large and complex wavelength (d) Complex and large wavelength
===== PAGE 22 =====

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239. [UNKNOWN] paper-ii

101. There are four dishes (i) a ladoo (spherical) with a radius of 4 cm; (ii) another ladoo with a radius of 8 cm; (iii) a pizza (disc) with a radius of 10 cm; and a pizza of same thickness with a radius of 20 cm. All the four dishes are simultaneously heated in an oven. Which one of them will be heated first? (a) Smaller ladoo will be heated first (b) Both the ladoos will be heated in the same time and prior to pizzas (c) Both the pizzas will be heated in the same time and prior to ladoos (d) All the four dishes will be heated in the same time

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240. [UNKNOWN] paper-ii

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241. [UNKNOWN] paper-ii

102. For a reversible isothermal process, the work done by the system is equal to: (a) Decrease in Helmholtz function (b) Increase in Helmholtz function (c) Decrease in Gibb's free energy (d) Increase in Gibb's free energy

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242. [UNKNOWN] paper-ii

102. For a reversible isothermal process, the work done by the system is equal to: (a) Decrease in Helmholtz function (b) Increase in Helmholtz function (c) Decrease in Gibb's free energy (d) Increase in Gibb's free energy

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243. [UNKNOWN] paper-ii

103. A resistor immersed in running water carries an electric current. Considering the resistor as the system, which of the following holds true? (a) $Q = 0$, $W = -ve$, $\Delta U = -ve$ (b) $Q = 0$, $W = -ve$, $\Delta U = +ve$ (c) $Q = 0$, $W = +ve$, $\Delta U = -ve$ (d) $Q = 0$, $W = +ve$, $\Delta U = +ve$

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244. [UNKNOWN] paper-ii

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245. [UNKNOWN] paper-ii

104. The difference of Helmholtz free energy F and Gibbs free energy G , (that is $F-G$) for an ideal gas sample of one

mole is: (a) RT (b) $-RT$ (c) T (d) $-T$

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

246. [UNKNOWN] paper-ii

104. The difference of Helmholtz free energy F and Gibbs free energy G , (that is $F-G$) for an ideal gas sample of one mole is: (a) RT (b) $-RT$ (c) T (d) $-T$

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Paper-II-GEOPHYSICS.pdf>

247. [UNKNOWN] paper-ii

105. What will be the radius of the nucleus with atomic mass 216, if nuclear radius parameter is $r_0 = 1.5$ fm? (a) 9.0 fm (b) 54.0 fm (c) 6.0 fm (d) 324 fm ===== PAGE 23 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

248. [UNKNOWN] paper-ii

105. What will be the radius of the nucleus with atomic mass 216, if nuclear radius parameter is $r_0 = 1.5$ fm? (a) 9.0 fm (b) 54.0 fm (c) 6.0 fm (d) 324 fm ===== PAGE 23 =====

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249. [UNKNOWN] paper-ii

106. Two spherical nuclei have mass numbers 216 and 64 with their radii a_1 and a_2 respectively. The ratio a_1/a_2 is: (a) 1.5 (b) 3 (c) 2.5 (d) 6

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geophysics.pdf>

250. [UNKNOWN] paper-ii

106. Two spherical nuclei have mass numbers 216 and 64 with their radii a_1 and a_2 respectively. The ratio a_1/a_2 is: (a) 1.5 (b) 3 (c) 2.5 (d) 6

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251. [UNKNOWN] paper-ii

107. Mass of a helium nucleus is 4.0028 amu, where as the mass of a proton is 1.00758 amu and that of neutron is 1.00897 amu. The binding energy of alpha particle in MeV unit will be: (1 amu) (a) 21.5 MeV (b) 0.0303 MeV (c) 28.21 MeV (d) 28.21×10^{-6} MeV

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252. [UNKNOWN] paper-ii

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253. [UNKNOWN] paper-ii

108. Which one of the following properties of the nuclear force is correct? (a) It depends on the charge of the nucleons (b) It depends on the directions of the spins of the nucleons (c) It does not depend on the directions of the spins of the nucleons (d) It depends on the mass of the nucleons

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254. [UNKNOWN] paper-ii

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255. [UNKNOWN] paper-ii

109. Which one of the following statements regarding β -decay is correct? (a) α -particles have a greater range than β -particles (b) Parity is not conserved in β -decay (c) The longest lived nuclei emit the most energetic particles (d) β -particles emitted by a radioactive element have same energy and range

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256. [UNKNOWN] paper-ii

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257. [UNKNOWN] paper-ii

110. Which one of the following statements is NOT correct? (a) A bright light yields more photoelectrons than a dim one of the same frequency (b) Red light produces faster photoelectrons than blue light (c) Pair production cannot occur in empty space (d) Compton shift is maximum for scattering angle of 180°

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258. [UNKNOWN] paper-ii

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259. [UNKNOWN] paper-ii

111. For pair production to take place γ -ray energy should be greater than: (a) 1.02 MeV (b) 2.56 MeV (c) 2.05 MeV (d) 0.78 MeV ===== PAGE 24 =====

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260. [UNKNOWN] paper-ii

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261. [UNKNOWN] paper-ii

112. A certain radioactive material has a life time of 30 days in which it reduces half to its amount that is $t_{1/2} = 30$ days. What will be its mean life? (a) 20.79 days (b) 15 days (c) 43.29 days (d) 231 days

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262. [UNKNOWN] paper-ii

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263. [UNKNOWN] paper-ii

113. Radon environmental hazard is associated with the radioactive decay of: (a) Rubidium (Rb-87) (b) Potassium (K-40) (c) Carbon (C-14) (d) Uranium (U-238)

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264. [UNKNOWN] paper-ii

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265. [UNKNOWN] paper-ii

114. Which one of the following statements regarding scintillation counter is correct? (a) It makes use of very fine drops of oil (b) It is not used for counting alpha particles (c) It uses a material which emits light when a charged particle strikes it (d) It cannot be used for detection of radiations but for scintillating the screen of a detector

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266. [UNKNOWN] paper-ii

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267. [UNKNOWN] paper-ii

115. Which one of the following instruments separates the characteristic γ -rays of ^{40}K , ^{238}U and ^{232}Th ? (a) Geiger Muller Counter (b) Scintillation Detector (c) Pulse-height analyzer (d) Radiation Counter

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268. [UNKNOWN] paper-ii

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269. [UNKNOWN] paper-ii

116. Which one of the following statement regarding Geiger-Muller Tube is correct? (a) It has a metallic casing filled with a high pressure gas (b) Both the ends of its tube are fixed with thin mica sheets (c) It has a metallic casing filled with low pressure gas (d) It can detect all radiations emerging out from a radioactive source ===== PAGE 25 =====

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270. [UNKNOWN] paper-ii

116. Which one of the following statement regarding Geiger-Muller Tube is correct? (a) It has a metallic casing filled with a high pressure gas (b) Both the ends of its tube are fixed with thin mica sheets (c) It has a metallic casing filled with low pressure gas (d) It can detect all radiations emerging out from a radioactive source ===== PAGE 25 =====

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271. [UNKNOWN] paper-ii

117. A particle of mass 10 gram is moving with velocity 10m/sec. What will be the De Broglie wavelength associated with the particle? (Plank's constant = 6.62×10^{-34} Js) (a) $6.62 \times 10^{-32}\text{m}$ (b) $6.62 \times 10^{-33}\text{m}$ (c) Zero (d) Infinity

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272. [UNKNOWN] paper-ii

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273. [UNKNOWN] paper-ii

118. Which of the following particle always obey Pauli exclusion principle? (a) All charged particles (b) All particles (c) All particles that have spin quantum numbers as multiples of $\frac{1}{2}$ (d) All particles that have spin quantum numbers as multiples of 1

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275. [UNKNOWN] paper-ii

119. The Balmer lines of hydrogen (transition to $n = 2$) span the visible frequency range. What is the maximum energy of the balmer line photons? (a) 3.4 eV (b) 1.9 eV (c) 13.6 eV (d) 7.5 eV

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276. [UNKNOWN] paper-ii

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277. [UNKNOWN] paper-ii

120. Which one of the following statement is NOT correct? (a) The longest wavelength line of Lyman series is the resonance line of hydrogen spectrum (b) Balmer series contains only those spectral lines of hydrogen atom, which occur in the visible part of the spectrum (c) In spectrum of helium, for each singlet energy state there is a corresponding triplet energy state (d) The sharp and diffuse series of spectra of alkali atoms have a common convergence limit

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278. [UNKNOWN] paper-ii

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